

CYBR-490 Capstone Project

Milestone 1 Report

Project Title:

AI Threat Intelligence Aggregator

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Changes to the Plan Document

Since the submission of the initial project plan, several updates have been made to the system design and development approach. These changes reflect progress in implementation and refinement of the system architecture.

The primary updates include:

1. Integration of Threat Intelligence Sources

The system now incorporates two initial threat intelligence sources:

- **CISA Known Exploited Vulnerabilities (KEV) Catalog**
- **MITRE ATLAS AI Threat Knowledge Base**

These sources were selected because they provide high-quality security intelligence and relevant information related to emerging threats.

2. Development of Data Ingestion Pipeline

The architecture has been updated to include a **Python-based data ingestion pipeline** responsible for retrieving and parsing threat intelligence data from external sources.

The ingestion layer standardizes data collected from different formats and prepares it for processing within the system.

3. Structured JSON Threat Dataset

Threat information is now stored in a **structured JSON format**. This format enables efficient processing and supports future integration with AI models and the dashboard interface.

The JSON dataset acts as the central data structure connecting the ingestion pipeline and the visualization layer.

4. Threat Categorization Framework

The project plan has been updated to include a threat classification structure consisting of three categories:

- **Hypothetical Threats**
- **Demonstrated Threats**
- **Active Exploitation**

This categorization helps security analysts better understand the maturity and severity of AI-related threats.

Updated Design Artifacts

System Requirements

The following requirements guide the development of the AI Threat Intelligence Aggregator.

Functional Requirements

The system must:

1. Collect threat intelligence data from external sources.
2. Parse and normalize threat information into a structured format.
3. Store threat information in a standardized JSON dataset.
4. Categorize threats into predefined threat groups.
5. Display threat information through a graphical dashboard.
6. Allow users to explore and filter threats.

Non-Functional Requirements

The system should:

- maintain reliable data processing pipelines
- support scalable threat ingestion
- ensure data integrity and validation
- present threat intelligence in a user-friendly interface

System Architecture

The system architecture consists of several layers responsible for collecting, processing, storing, and displaying threat intelligence data.

Architecture Overview

```
graph TD; A["Threat Sources  
(CISA KEV, MITRE ATLAS)"] --> B["Data Ingestion Layer  
(Python scripts retrieve and parse data)"]; B --> C["Data Processing Pipeline  
(cleaning, normalization, schema mapping)"]; C --> D["Threat Categorization  
(Hypothetical / Demonstrated / Active Exploitation)"]; D --> E["Structured JSON Dataset  
(paper_summaries.json)"]; E --> F["Dashboard GUI  
(threat visualization interface)"];
```

The diagram illustrates the system architecture as a vertical flow of seven layers, connected by downward-pointing arrows. The layers are: Threat Sources (CISA KEV, MITRE ATLAS), Data Ingestion Layer (Python scripts retrieve and parse data), Data Processing Pipeline (cleaning, normalization, schema mapping), Threat Categorization (Hypothetical / Demonstrated / Active Exploitation), Structured JSON Dataset (paper_summaries.json), and Dashboard GUI (threat visualization interface).

Implementation Progress

At this stage of development, several core components of the system have been implemented or are currently under development.

Data Ingestion

The project currently includes ingestion mechanisms for the following threat intelligence sources:

- **CISA KEV Catalog**
- **MITRE ATLAS database**

Python scripts retrieve relevant threat information and extract key fields from these sources.

Data Processing Pipeline

The processing pipeline performs several tasks:

- cleaning raw data
- normalizing threat fields
- structuring records into a consistent schema

This pipeline ensures that data from different sources can be processed consistently.

JSON Threat Data Structure

Threat records are stored in a structured JSON format. Each record contains information describing a threat entry.

Example JSON record:

```
</> JSON

{
  "title": "Prompt Injection Attack",
  "source": "MITRE ATLAS",
  "summary": "Attack manipulating LLM prompts to influence model output",
  "category": "Active Exploitation",
  "risk_level": "High",
  "url": "https://example-source"
}
```

Updated Timeline

The following timeline summarizes completed tasks and remaining work.

Tasks Completed or In Progress

Task	Status
Data ingestion framework	In Progress
Data processing pipeline	In Progress
CISA KEV integration	In Progress
MITRE ATLAS integration	In Progress

Next Development Phase

The next stage of development will focus on completing the integration between the backend data pipeline and the graphical dashboard interface.

Key objectives include:

- finalizing threat categorization
- improving the dashboard interface
- integrating Python modules into a unified application
- improving project organization and code structure

Conclusion

This milestone represents significant progress toward building the AI Threat Intelligence Aggregator. The development team has established the core data ingestion and processing pipeline and has begun integrating real-world threat intelligence sources.

Future work will focus on completing system integration, improving the dashboard interface, and expanding the threat dataset to support advanced threat analysis.